

- by Older People with Functional and Social Frailty: Ageing in Place in Deprived Areas of Italy*. PMC. Retrieved March 10, 2026, from <https://pmc.ncbi.nlm.nih.gov/articles/PMC9691100/>
- Migration Survey Report, Malaysia, 2024*. (2025, December 18). Department of Statistics Malaysia. Retrieved February 7, 2026, from <https://www.dosm.gov.my/portal-main/release-content/migration-survey-report-malaysia-2024>
- Patel, S., Hyung Park, Paolo Bonato, Leighton Chan, & Mary Rodgers. (2012, April 20). *A review of wearable sensors and systems with application in rehabilitation*. Journal of NeuroEngineering and Rehabilitation. Retrieved February 11, 2026, from <https://jneuroengrehab.biomedcentral.com/articles/10.1186/1743-0003-9-21>
- Pew Research Center. (2025, November 20). *Demographics of Mobile Device Ownership and Adoption in the United States*. Pew Research Center. Retrieved February 24, 2026, from <https://www.pewresearch.org/internet/fact-sheet/mobile/>
- Rashidi, P., & Mihailidis, A. (2013, May). *A Survey on Ambient-Assisted Living Tools for Older Adults*. IEEE.org. Retrieved March 7, 2026, from <https://ieeexplore.ieee.org/document/6399501>
- Statista. (2025, November 29). *Share of children in total population in Malaysia from 2015 to 2024*. Statista. Retrieved February 7, 2026, from https://www.statista.com/statistics/678083/malaysia-children-as-a-percentage-of-the-population/#:~:text=Table_title:%20Share%20of%20children%20in%20total%20population,%7C%20Share%20of%20total%20population:%2024.9%25%20%7C
- Venkatesh, V. (2003). *User acceptance of information technology: Toward a unified view*. Website title: MIS Quarterly. MIS Quarterly. Retrieved February 11, 2026, from <https://doi.org/10.2307/30036540>
- Vlaev, I., King, D., Dazil, A., & Dolan, P. (2019, August 7). *Changing health behaviors using financial incentives: a review from behavioral economics*. PMC. Retrieved March 10, 2026, from <https://pmc.ncbi.nlm.nih.gov/articles/PMC6686221/>
-

- Wang, R. (2014, September 13). *StudentLife: Assessing mental health, academic performance and behavioral trends of college students using smartphones*. UbiComp '14: Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing. Retrieved February 11, 2026, from <https://dl.acm.org/doi/10.1145/2632048.2632054>
- Wang, Z., Wang, Y., Zeng, Y., Su, J., & Li, Z. (2025, May 23). *An investigation into the acceptance of intelligent care systems: an extended technology acceptance model (TAM)*. Scientific Reports. Retrieved March 10, 2026, from <https://www.nature.com/articles/s41598-025-02746-w?>
- World Health Organization. (2015). *World Health Organization*. World report on ageing and health. Retrieved February 24, 2026, from <https://iris.who.int/server/api/core/bitstreams/9d69518e-21e4-48c4-980d-8f6125d13aa9/content>
- World Population Prospects 2022: Summary of Results*. (2022). the United Nations. Retrieved February 11, 2026, from <https://www.un.org/development/desa/pd/>
- Stanley, S., Hayes, J. A., Mogan, C., Lloyd-Williams, M., Nwosu, A. C., Vasilica, C. M., & Higginbotham, K. (2026). Digital Legacy in Palliative Care and End-of-Life Care Planning: A Scoping Review. *BMJ Supportive & Palliative Care*, spcare-2026-006205. <https://doi.org/10.1136/spcare-2026-006205>
- Supabase. (2026, July 31). *Serving Assets from Storage | Supabase Docs*. Supabase Docs. Supabase. <https://supabase.com/docs/guides/storage/serving/downloads?>
- Malaysian Meteorological Department Web Service API. (n.d.). In *api.met.gov.my*. Retrieved August 2, 2026, from <https://api.met.gov.my/>
- Open-Meteo.com. (2026). In *Open-meteo.com*. <https://open-meteo.com/en/docs?>

Project Title

References

Android Developers. (n.d.). WorkManager overview. Retrieved August 3, 2026, from <https://developer.android.com/develop/background-work/background-tasks/persistent/getting-started>

Brevo. (n.d.). Send a transactional email. Retrieved August 3, 2026, from <https://developers.brevo.com/docs/send-a-transactional-email>

Flutter. (n.d.). Guide to app architecture. Retrieved August 3, 2026, from <https://docs.flutter.dev/app-architecture/guide>

Flutter. (n.d.). local_auth package. Retrieved August 3, 2026, from https://pub.dev/packages/local_auth

Government of Malaysia. (n.d.). Weather API. Retrieved August 3, 2026, from <https://developer.data.gov.my/realtime-api/weather>

Supabase. (n.d.). Auth. Retrieved August 3, 2026, from <https://supabase.com/docs/guides/auth>

Supabase. (n.d.). Edge Functions. Retrieved August 3, 2026, from <https://supabase.com/docs/guides/functions>

Supabase. (n.d.). Row Level Security. Retrieved August 3, 2026, from <https://supabase.com/docs/guides/database/postgres/row-level-security>

Supabase. (n.d.). Scheduling Edge Functions. Retrieved August 3, 2026, from <https://supabase.com/docs/guides/functions/schedule-functions>

Twilio. (n.d.). Message resource. Retrieved August 3, 2026, from <https://www.twilio.com/docs/messaging/api/message-resource>

Flutter. (2026). Accessibility. Retrieved August 5, 2026, from <https://docs.flutter.dev/ui/accessibility>

OWASP Foundation. (n.d.). MASWE-0006: Sensitive data stored unencrypted in private storage locations. Retrieved August 5, 2026, from <https://mas.owasp.org/MASWE-0006/>

Google Play. (n.d.). Use of SMS or Call Log permission groups. Retrieved August 5, 2026, from <https://support.google.com/googleplay/android-developer/answer/10208820?hl=en>

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Appendices

Appendices

Appendix A: User Guide

Account and Setup

Install or launch Ethernacare, create an account or continue with Google, complete email confirmation where required, accept the Terms and Conditions, complete the profile, verify the user phone and add a verified primary trusted contact. The first completed setup opens an eight-step guide covering Home, History, Contacts, Rewards, AI Guidance, Legacy Planning, Profile and safety boundaries. The guide can be skipped and replayed later from Profile or Settings. No shared demonstration password is embedded in the application.

Daily Operation

Open Home and review Oren's status and the Safety Monitor. Tap Oren when the rolling threshold allows a check-in. Feed and Play change Oren care state but do not count as a safety heartbeat. Use History to inspect successful timestamps, Contacts to manage the verified safety network, Rewards to collect virtual items and Profile to change threshold, escalation and biometric settings. Open Profile > Settings to control local reminders, Oren sounds, text size and reduced motion, replay the feature guide or restore defaults.

Emergency and Legacy Functions

Use SOS only when a trusted contact should be notified; location is included only when permission and coordinates are available. Use the 999 action only for a real emergency because it opens an explicit call flow. In Legacy Planning, save funeral preferences, safe notes and supported private files, choose the authorised existing primary contact and enable release consent. Never store passwords, PINs, OTPs, recovery phrases or account credentials in Legacy Notes.

Appendix B: Developer Guide

Required Software

Flutter/Dart compatible with the project SDK constraint, Android Studio or Visual Studio Code, Android SDK for mobile builds, Visual Studio with Desktop development with C++ for Windows builds, Git, Node.js/npm for the Supabase CLI, and access to the configured Supabase project.

Backend Setup

Apply SQL migrations in supabase/migrations in filename order. Deploy all functions in supabase/functions, configure encrypted Twilio, email, AI and OTP secrets, verify authentication redirect URLs and providers, confirm private Storage policies, and inspect the 15-minute threshold and midnight Legacy Cron jobs. Confirm that production builds omit ENABLE_SAFETY_TESTS, set the final Android application identifier and include only approved platform permissions. Do not commit secret values or include them in screenshots.

Verification

Run flutter analyze, flutter test --coverage and the required release builds. The latest recorded automated run contains 138 passing tests with 2,515 of 8,208 executable lines covered (30.64 percent). Confirm database migration parity and Edge Function health. Complete physical Android checks for notification permission, background/resume behaviour, biometric success/cancel/retry, GPS capture and real SMS/email receipt. Record provider message IDs and errors without exposing tokens.